

# PROJECT OVERVIEW & DATA INFRASTRUCTURE

## 1.1 Project Title:

**Comprehensive Business Intelligence & Performance Report for CoreTech Innovation.**

## 1.2 Dataset Overview:

The dataset consists of operational and financial records from CoreTech Innovation, tracking performance across various dimensions:

- **Total Records:** 50 Active Clients/Projects.
- **Key Fields:** Client\_ID, Client\_Name, Client\_City, Client\_Services, Project\_Budget, Project\_Status, Payment\_Status, Lead\_Sources, Start\_Date, End\_Date.

## 1.3 Tools Used:

- **Excel / WPS Office:** Used for initial data assessment, structure handling, and manual validation.
- **SQL (DB Fiddle):** Executed advanced relational queries to isolate structural metrics and answer deep data-driven business questions.
- **Power BI Desktop:** Implemented to model data, develop dynamic KPI cards, establish interactive filters/slicers, and visualize data trends.

## 1.4 Data Cleaning Summary:

Before visualization, the raw data underwent a rigorous data cleaning pipeline to ensure maximum analytical accuracy:

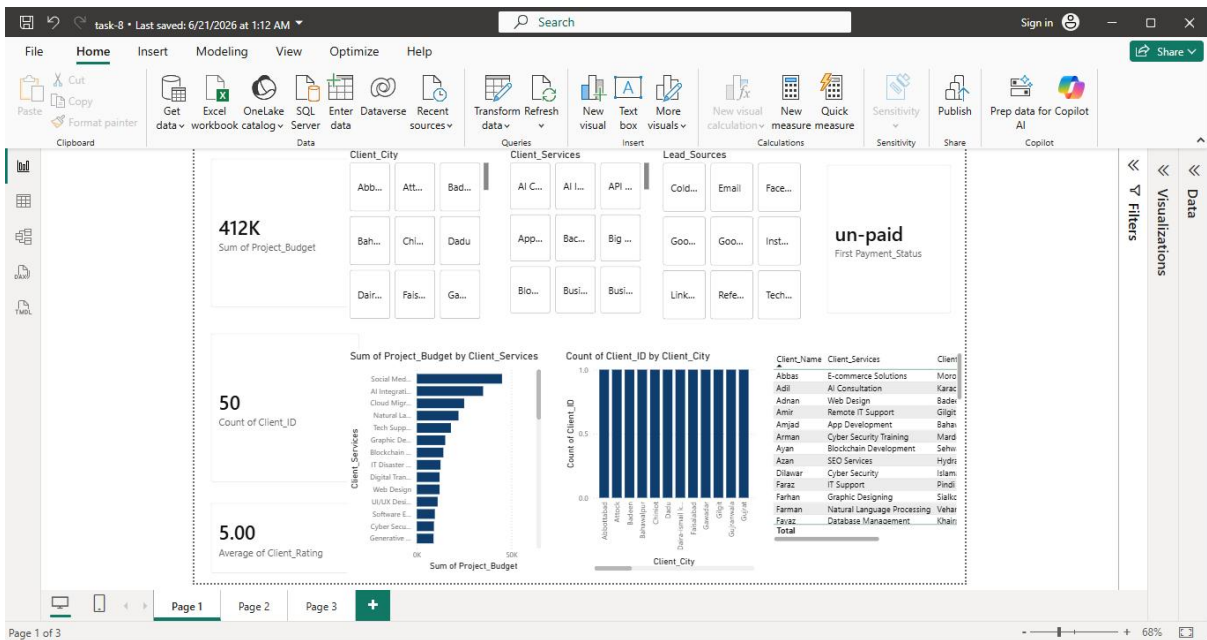
- **Duplicate Elimination:** Scanned and removed redundant duplicate entries to ensure unique Client IDs.
- **Standardization:** Cleaned and uniformized categorical texts across columns such as Client\_Services and Lead\_Sources.
- **Data Type Formatting:** Adjusted currency values to numbers (Project\_Budget), formatted temporal markers (Start\_Date, End\_Date), and verified proper statistical categories.

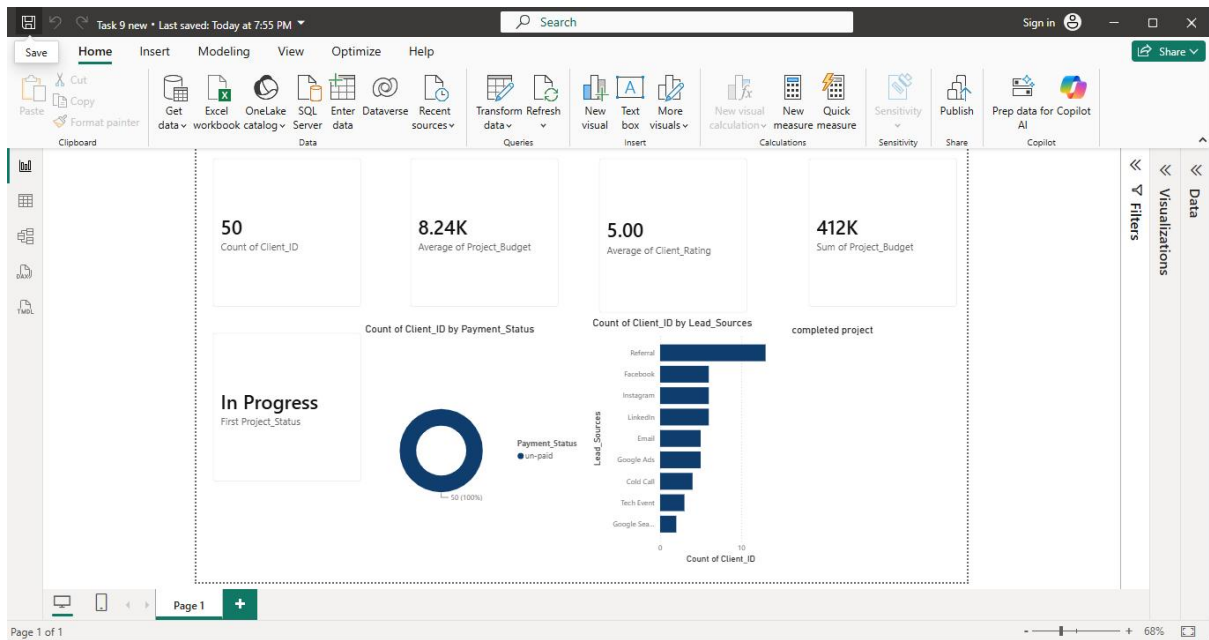
# VISUALIZATION & KPI SUMMARY

## 2.1 Charts Summary

The analytical dashboard was structured around distinct, meaningful visuals to convey performance:

- **Clustered Bar Chart:** Organized Sum of Project\_Budget by Client\_Services to rank top operational services vertically.
- **Column Chart:** Plotted client distribution across various regional cities (Count of Client\_ID by Client\_City).
- **Donut Chart / Visual Mix:** Mapped out structural parameters such as lead performance metrics and overall project statuses.





## 2.2 KPI Summary Table:

Below is the validated list of core operational metrics derived from the analytical data model:

| KPI Metric Name         | Captured Value    | Strategic Business Definition                                         |
|-------------------------|-------------------|-----------------------------------------------------------------------|
| Total Revenue Pipeline  | 412,200 PKR       | Total financial volume locked within current project portfolios.      |
| Total Clients           | 50 Active Clients | Count of unique active corporate relationships.                       |
| Average Project Budget  | 8.24K PKR         | Mean financial size expected per contract execution.                  |
| Average Rating          | 5.00 / 5.00       | Ultimate metric tracking client satisfaction and delivery quality.    |
| Completed Projects      | 0 (Blank)         | Projects fully signed off and delivered to date.                      |
| Active/Pending Projects | 50                | Ongoing operations actively managed by the team.                      |
| Primary Lead Source     | Cold Call         | Top alphabetical data marker for client acquisition channels.         |
| Initial Payment Status  | Un-paid           | Current baseline state of financial collections across active tracks. |

## **BUSINESS INSIGHTS, PROBLEMS, & STRATEGI** **RECOMMENDATIONS**

### **3.1 Key Business Insights (8 Insights Required)**

- 1) **High-Value Portfolio Revenue:** CoreTech Innovation holds a robust gross pipeline of 412,200 PKR locked in active engagements.
- 2) **Exceptional Delivery Standards:** A flawless 5.0/5.0 Average Client Rating reflects extraordinary service standards and client satisfaction.
- 3) **Heavy Operational Load:** 100% of the active portfolio is classified as In Progress, meaning the internal workforce is operating at maximum capacity.
- 4) **Zero Milestone Realization:** No projects have reached Completed status, highlighting a potential bottleneck in the final sign-off phase.
- 5) **Critical Cash Flow Deficit:** Every active project is categorized as un-paid, creating a high operational risk due to lack of immediate liquidity.
- 6) **Top Revenue-Generating Service:** Digital transformation tracks, led strongly by *Social Media Services*, command the highest share of the project budget.
- 7) **Widespread Regional Footprint:** The business is diversified across multiple client locations (including cities like *Abbottabad, Badin, Chiniot*, etc.), reducing geographical risk.
- 8) **Heavy Lead Dependence:** Inbound acquisition relies significantly on baseline outreach frameworks such as manual sourcing and cold calls.

### **3.2 Business Problems Identified:**

- 1) **The Cash Flow Trap:** Operating with zero upfront cash collection puts intense strain on day-to-day business finances.
- 2) **Delivery Bottleneck:** Managing 50 active tracks simultaneously without structured, staggered rollouts stretches resource optimization.
- 3) **Quality Control & Retention Risk:** Maintaining a flawless 5.0 client rating across 50 simultaneous active projects poses a critical operational risk, as resource over-extension can lead to delivery delays and a decline in service quality.
- 4) **Client Retention & Quality Risk:** Perfect 5.0 rating maintain rakhna 50 active projects par bohot mushkil hota hai, kyunki workload barhne se execution quality kharab hone ka khatra barh jata hai.

### **3.3 Strategic Recommendations (5 Recommendations Required)**

- 1) Deploy Milestone-Based Billing:** Immediately transition away from a 100% unpaid model. Implement a 30-40-30 collection structure (Upfront, Mid-way, Completion) to secure immediate working capital.
- 2) Establish Project Staging Pipelines:** Introduce rigorous Agile tracking to break the "In Progress" logjam. Group projects into phased delivery cycles to push them into "Completed" systematically.
- 3) Automate Acquisition Channels:** Broaden the lead funnel. Pivot from manual cold calling to inbound digital marketing architectures to drive higher-converting corporate queries.
- 4) Premium Service Packaging:** Capitalize on the perfect 5.0 client rating. Package high-value solutions (like Social Media and AI tracks) into premium enterprise tiers with a 15-20% higher budget baseline.
- 5) Expand Operational Delivery Teams:** If bandwidth constraints are slowing down final project completions, reinvest projected revenue into hiring specialized contract execution experts.

### **3.4 Conclusion**

CoreTech Innovation showcases exceptional quality execution and market demand, evidenced by a perfect client rating and a strong pipeline. However, translating pipeline value into actual liquid cash flow through milestone collection models and structural phase rollouts remains the key requirement for sustainable scaling.